UNIVERSITY OF THE PEOPLE

MATH 1280-01 - AY2026-T4

Assignment Activity Unit 2

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Independent, Mutually Exclusive Events, Venn Diagrams, Tree Diagrams, and Probability Calculations

Introduction

Probability is a foundational branch of mathematics that enables us to measure and quantify uncertainty. Understanding core probability concepts is essential not only in academic contexts but also in real-world decision-making, data analysis, and machine learning. This assignment explores key probability terminology, contingency-based calculations using Venn and tree diagrams, and probability computations involving replacement and non-replacement scenarios, all grounded in activities designed for Greenfield High School students.

Activity 1: Key Probability Terminology

The following definitions are contextualized within the activities described: Group A examines outcomes from rolling a six-sided die categorized as even or odd, while Group B explores flipping a coin followed by rolling a six-sided die.

1. Experiment

An experiment is a process or procedure that is carried out under controlled or defined conditions to observe and record outcomes whose results are uncertain in advance (Illowsky & Dean, 2023). In Group A, the experiment is rolling a standard six-sided die and categorizing the result as even or odd. In Group B, the experiment consists of two sequential actions: first flipping a coin and then rolling a six-sided die.

2. Sample Space

The sample space (S) is the complete set of all possible outcomes of an experiment (Ross, 2019). For Group A, the sample space is S = {1, 2, 3, 4, 5, 6}, representing all faces of the die. For Group B, the sample space is the set of all ordered pairs: S = {(H,1), (H,2), (H,3), (H,4), (H,5), (H,6), (T,1), (T,2), (T,3), (T,4), (T,5), (T,6)}, yielding 12 equally likely outcomes.

3. Event

An event is any subset of the sample space a specific collection of one or more outcomes of interest (Illowsky & Dean, 2023). In Group A, rolling an even number defines an event: E = {2, 4, 6}. Similarly, rolling an odd number defines another event: O = {1, 3, 5}. In Group B, an event could be "getting Heads followed by a number greater than 4," represented as {(H,5), (H,6)}.

4. Outcome

An outcome is a single specific result of an experiment — the most elementary unit within a sample space (Ross, 2019). In Group A, rolling the number 3 is one outcome. In Group B, obtaining Tails on the coin and rolling a 4 on the die is one distinct outcome, written as (T, 4).

5. Probability

Probability is a numerical measure of the likelihood that a specific event will occur, expressed as a value between 0 (impossible) and 1 (certain), inclusive. It is formally defined as:

$$P(A) = (\text{Number of favorable outcomes}) / (\text{Total number of outcomes in sample space})$$

In Group A, the probability of rolling an even number is P(E) = 3/6 = 0.5. In Group B, the probability of getting Heads and rolling a 5 is P(H, 5) = 1/12 ≈ 0.083 (Illowsky & Dean, 2023).

6. Mutually Exclusive Events

Two events are mutually exclusive (also called disjoint) if they cannot occur simultaneously — their intersection is empty (Ross, 2019). Formally: P(A ∩ B) = 0. In Group A, the events "rolling an even number" and "rolling an odd number" are mutually exclusive because no number can be both even and odd at the same time. For mutually exclusive events, the Addition Rule simplifies to:

$$P(A \cup B) = P(A) + P(B)$$

Thus, P(Even ∪ Odd) = 0.5 + 0.5 = 1, which correctly confirms these two events cover the entire sample space.

7. Independent Events

Two events are independent if the occurrence of one does not influence the probability of the other (Illowsky & Dean, 2023). Formally, A and B are independent if and only if:

$$P(A \cap B) = P(A) \times P(B)$$

In Group B, flipping a coin and rolling a die are independent events because the coin result has no effect on which face the die shows. For example, P(Heads) = 1/2 and P(rolling a 3) = 1/6, so P(Heads and 3) = (1/2) × (1/6) = 1/12, consistent with the 12-element sample space.

Activity 2: Venn Diagram and Tree Diagram

A survey of 150 students was conducted regarding participation in two workshops: Leadership (L) and Team Building (T). The data is organized in the contingency table below:

Workshops	Leadership (L)	No Leadership (L')	Total
Team Building (T)	50	30	80
No Team Building (T')	20	50	70
Total	70	80	150

Part A: Venn Diagram

Based on the contingency table, the Venn diagram values are derived as follows:

Both Leadership AND Team Building (L ∩ T) = 50 students

Leadership ONLY (L ∩ T') = 20 students

Team Building ONLY (T ∩ L') = 30 students

Neither Leadership nor Team Building (L' ∩ T') = 50 students

Verification: 50 + 20 + 30 + 50 = 150 students ✓

Venn Diagram – Workshop Participation (n = 150)

Universal Set U = 150 students		
L only 20	L ∩ T 50	T only 30
Neither (L' ∩ T') = 50		

Part B: Tree Diagram

The tree diagram below illustrates all conditional probability paths for student workshop participation. Probabilities are expressed as fractions over 150 students.

Step 1 – First Branch (Leadership status):

P(L) = 70/150 = 7/15 ≈ 0.467

P(L') = 80/150 = 8/15 ≈ 0.533

Step 2 – Second Branch (Team Building given Leadership status):

P(T | L) = 50/70 = 5/7 ≈ 0.714

P(T' | L) = 20/70 = 2/7 ≈ 0.286

P(T | L') = 30/80 = 3/8 = 0.375

P(T' | L') = 50/80 = 5/8 = 0.625

Tree Diagram – Workshop Participation Probabilities

Branch 1 (Leadership)	Probability	Branch 2 (Team Building)	Conditional Prob.	Joint Probability	Count
L (Leadership)	70/150 = 0.467	T (Team Building)	50/70 = 0.714	P(L ∩ T) = 50/150 ≈ 0.333	50
L (Leadership)	70/150 = 0.467	T' (No Team Building)	20/70 = 0.286	P(L ∩ T') = 20/150 ≈ 0.133	20
L' (No Leadership)	80/150 = 0.533	T (Team Building)	30/80 = 0.375	P(L' ∩ T) = 30/150 = 0.200	30
L' (No Leadership)	80/150 = 0.533	T' (No Team Building)	50/80 = 0.625	P(L'∩ T') = 50/150 ≈ 0.333	50
			TOTAL	150/150 = 1.000	150

Activity 3: Probability Calculations – Green and Yellow Balls

A box contains 8 green balls and 4 yellow balls, giving a total of 12 balls. The task is to calculate the probability of drawing a green ball first, followed by a yellow ball, under two different conditions.

Case 1: With Replacement (Independent Events)

When the first ball is replaced before drawing the second, the total number of balls remains 12 for both draws. The two draws are independent events.

Given:

Total balls (N) = 12

Green balls (G) = 8

Yellow balls (Y) = 4

Formula (Multiplication Rule for Independent Events):

$$P(G \text{ then } Y) = P(G) \times P(Y)$$

Step-by-step calculation:

P(G) = 8/12 = 2/3

P(Y) = 4/12 = 1/3 [sample space unchanged because ball was replaced]

P(G then Y) = (8/12) × (4/12) = 32/144 = 2/9 ≈ 0.2222

Conclusion: The probability of drawing a green ball followed by a yellow ball with replacement is 2/9 ≈ 0.222 or approximately 22.2%.

Case 2: Without Replacement (Dependent Events)

When the first ball is NOT replaced before drawing the second, the total number of balls decreases from 12 to 11 for the second draw. The two events are now dependent.

Given:

Total balls (N) = 12

Green balls (G) = 8

Yellow balls (Y) = 4

Formula (Multiplication Rule for Dependent Events):

$$P(G \text{ then } Y) = P(G) \times P(Y \mid G)$$

Step-by-step calculation:

P(G) = 8/12 [8 green balls out of 12 total]

P(Y | G) = 4/11 [4 yellow balls remain out of 11 total balls after removing one green]

P(G then Y) = (8/12) × (4/11) = 32/132 = 8/33 ≈ 0.2424

Conclusion: The probability of drawing a green ball followed by a yellow ball without replacement is 8/33 ≈ 0.242 or approximately 24.2%.

Comparison and Explanation

The probability in Case 2 (without replacement, 8/33 ≈ 0.242) is slightly higher than in Case 1 (with replacement, 2/9 ≈ 0.222). This is because removing a green ball in Case 2 slightly increases the relative proportion of yellow balls remaining in the pool, making yellow more likely to be drawn second. This distinction illustrates why understanding dependence versus independence is critical in probability modeling and statistical inference (Ross, 2019).

Conclusion

This assignment examined fundamental probability concepts within the engaging context of Greenfield High School activities. The definitions of experiment, sample space, outcome, event, probability, mutually exclusive events, and independent events were grounded in realistic scenarios involving dice and coins. The Venn and tree diagrams for the workshop survey provided clear visual representations of joint and conditional probabilities. Finally, the ball-drawing problems illustrated how replacement or its absence fundamentally changes the nature of probability calculations, distinguishing independent from dependent events. Mastery of these

concepts forms the essential foundation for advanced topics such as Bayesian inference, hypothesis testing, and machine learning algorithms.

References

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